Sample.txt:

```
sample.txt
1    Hello my name is Atharv Pawar!
2
3    I am pursuing mca dfrom imcc
```

Q1. Write a program to read a file and count how many vowels are present in its content.

```
filename = "D:/abc/sample.txt"

vowels = "aeiouAEIOU"
count = 0

f = open(filename, 'r')
text = f.read()

for char in text:
    if char in vowels:
        count += 1

print(count)
```

Output:

```
PS D:\abc> python -u "d:\abc\vowel_count.py"
9
PS D:\abc>
```

Q2. Read all the lines from a file into a list. Print the total number of lines and display the 2nd and 4th line

```python
filename = "D:/abc/sample.txt"

f = open(filename, 'r')
lines = f.readlines()
f.close()

print(f"Total lines: {len(lines)}")

if len(lines) >= 2:
    print(f"2nd line: {lines[1]}")

if len(lines) >= 4:
    print(f"4th line: {lines[3]}")
```

Output:

```
PS D:\abc> python -u "d:\abc\readline.py"
Total lines: 3
2nd line:
```

Q3. Take a sentence from the user, split it into words and store them in a list

print the longest word.

```python
sentence = input("Enter a sentence: ")

words = sentence.split()

longest = words[0]

for word in words:
    if len(word) > len(longest):
        longest = word

print(f"Words list: {words}")
print(f"Longest word: {longest}")
```

Output:

```
PS D:\abc> python -u "d:\abc\split_sentence_print_longest_word.py"
Enter a sentence: I am Atharv
Words list: ['I', 'am', 'Atharv']
Longest word: Atharv
PS D:\abc>
```

Q4. Read a file split its content into words and store them in a list.

i. **int the total number of words**

```python
filename = "D:/abc/sample.txt"

f = open(filename, 'r')
text = f.read()
f.close()

words = text.split()

print(f"Total words: {len(words)}")
print(words)
```

Output:

```
PS D:\abc> python -u "d:\abc\totword.py"
Total words: 12
['Hello', 'my', 'name', 'is', 'Atharv', 'Pawar!', 'I', 'am', 'pursuing', 'mca', 'dfrom', 'imcc']
PS D:\abc>
```

ii. **Print all unique words**

```
filename = "D:/abc/sample.txt"

f = open(filename, 'r')
text = f.read()
f.close()

words = text.split()
unique_words = list(set(words))

print(f"Unique words: {unique_words}")
print(f"Total unique: {len(unique_words)}")
```

Output:

```
['Hello', 'my', 'name', 'is', 'Atharv', 'Pawar!', 'I', 'am', 'pursuing', 'mca', 'dfrom', 'imcc']
PS D:\abc> python -u "d:\abc\unique.py"
Unique words: ['am', 'is', 'Atharv', 'dfrom', 'name', 'imcc', 'Pawar!', 'my', 'I', 'pursuing', 'Hello', 'mca']
Total unique: 12
PS D:\abc>
```